

Pre-Algebra Final Test

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show all work.
- iii) Draw a rectangle around your answers.
- iv) Rewrite the equation whenever you perform an operation on both sides to preserve the result of the previous step.

Graph the following equation. Show all work and identify the points you use to graph the equation. Label your graph properly.

① $x - 4y = -5$

Graph the following inequality. Show all work and identify the points you use to graph the inequality. Label the graph properly.

② $y > \frac{x}{2} + 1$

Find the algebraic solutions and the graphical solutions of the following inequalities.

③ $x \geq 1 + 2x - 3$ ④ $-\frac{y}{3} + \frac{5y}{6} < 2y - 1$

Solve.

⑤ $-5(z - 3) + 2(3z + 2) = 10z - 3(-z + 1)$

$$\textcircled{6} 4.6x = 5.2(x-1) + 3.7 \quad \textcircled{7} -y + 7y + 4 = -3y - y$$

$$\textcircled{8} -\frac{z}{2} + \frac{z}{4} - 1 - \frac{2z}{5} = 4 - 6 \quad \textcircled{9} 2x - 10x = -6x + x - 3$$

Evaluate.

$$\textcircled{10} 7y^2 - 4y - 2(3y-4)^2, y=2 \quad \textcircled{11} -z^2 + \frac{z}{2} + z^3, z=-1$$

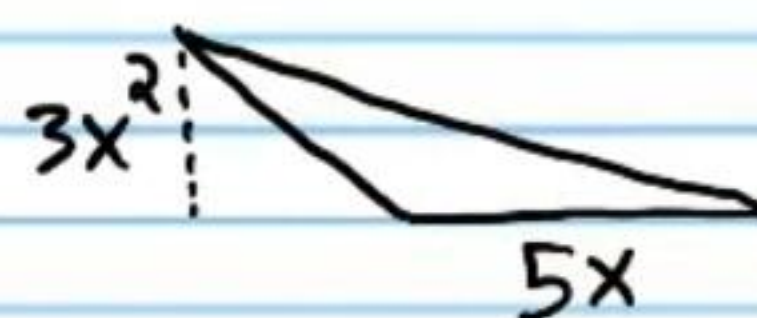
$$\textcircled{12} 2 - \frac{x+4}{1-3x} + 3(x-10), x=0 \quad \textcircled{13} \text{Simplify the quotient below.}$$

$$-\frac{5x^5y}{2z} \div \left(-\frac{3xz}{2y^4}\right)$$

- ⑭ Evaluate your answer for the previous problem if $x = -1$, $y = 1$, and $z = -2$.
 ⑮ Find A if $A = P + Prt$, $P = 350$, $r = 0.1$, and $t = 8$.

- ⑯ Find the area of a rectangle with length $\frac{7x^5}{8y^2z^3}$ and width $\frac{4x}{5yz^9}$.

- ⑰ Find the area of the triangle below.



- ⑱ a) What percent of 3.6 is 2.0?
 b) Simplify.

$$\frac{20a^3b^5c^{10}}{36b^3c^4a}$$

- ⑲ A factory makes 213 stuffed bears every hour. After the machines are upgraded, the factory makes 263 bears every hour. What was the percent increase in bears made per hour? Round to the nearest hundredth of a percent.

- ⑳ Find the volume of a rectangular prism with length $3x$, width $4.7x^3$, and height $2.87x$.

- ㉑ Find the width of a rectangle with length 3 inches and perimeter 18 inches.

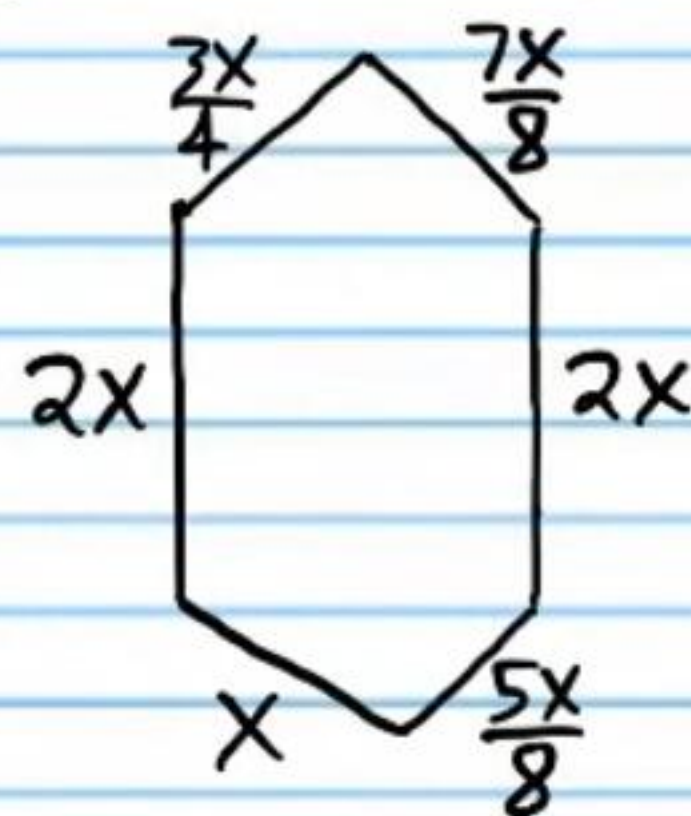
(22) Your friend loans you \$380 and charges simple interest yearly. After two years, your debt grows to \$395.20. Assuming you kept all the money, what is the interest rate of the loan?

(23) The sum of three consecutive odd integers is 273. Identify these integers.

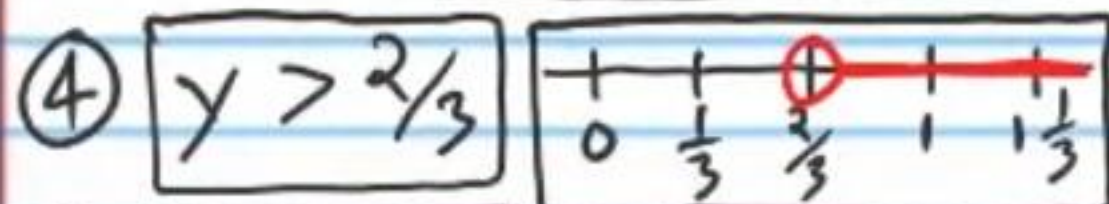
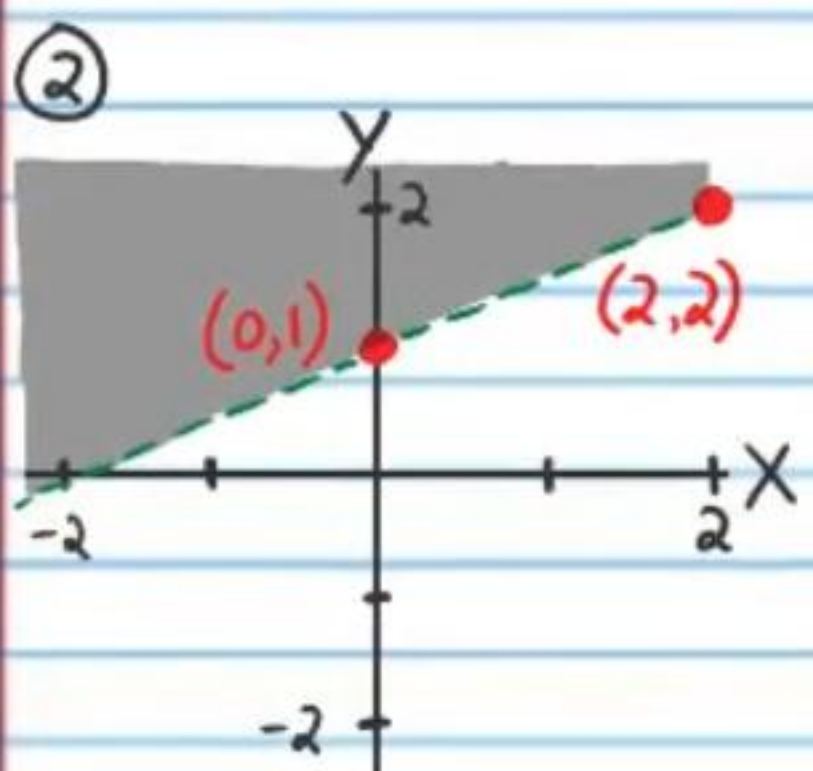
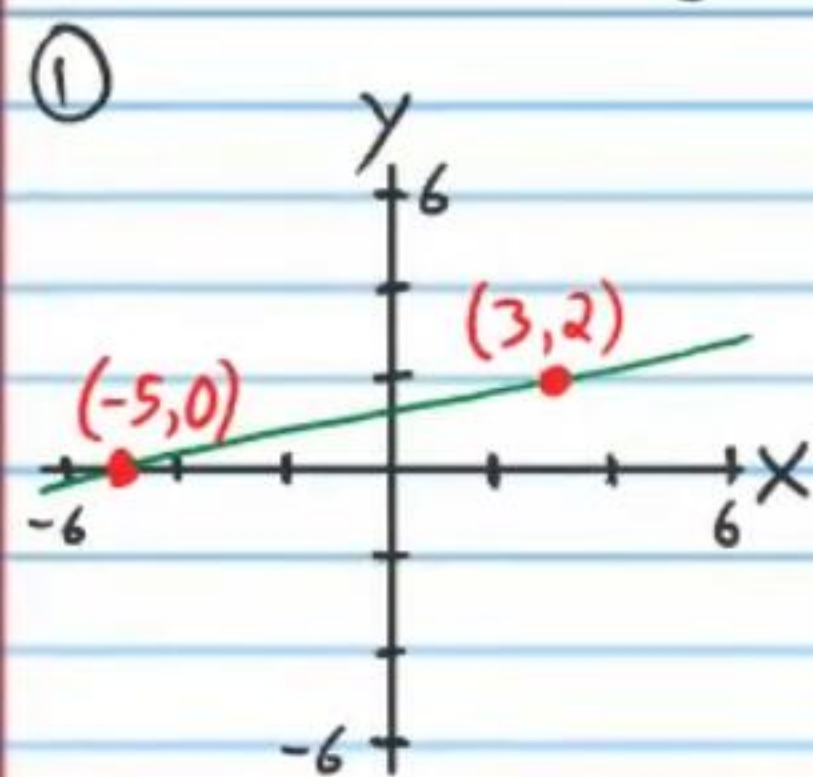
(24) a) 36% of what number is 400?
b) Simplify.

$$\frac{-15a^7c^3}{-5a^{10}bc^2}$$

(25) Find the perimeter of the hexagon below.



Pre-Algebra Final Test Answers



⑤ $z = 1\frac{1}{6}$

⑥ $x = 2.5$

⑦ $y = -\frac{2}{5}$

⑧ $z = \frac{20}{13}$

⑨ $x = 1$

⑩ 12

⑪ $-\frac{5}{2}$

⑫ -32

⑬ $\frac{5x^4y^5}{3z^2}$

⑭ $\frac{5}{12}$

⑮ 630

⑯ $\frac{7x^6}{10y^3z^{12}}$

⑰ $\frac{15x^3}{2}$

⑱ a) 555.5% b) $\frac{5a^2b^2c^6}{9}$

⑲ 23.47%

⑳ $40.467x^5$

㉑ 6 in

㉒ 2%

㉓ 89, 91, 93

㉔ a) I, III, T b) $\frac{3b^6c}{a^9}$

㉕ $\frac{29x}{4}$